Task: Prepare the Data for Visualization

Project: Heart Disease Analysis
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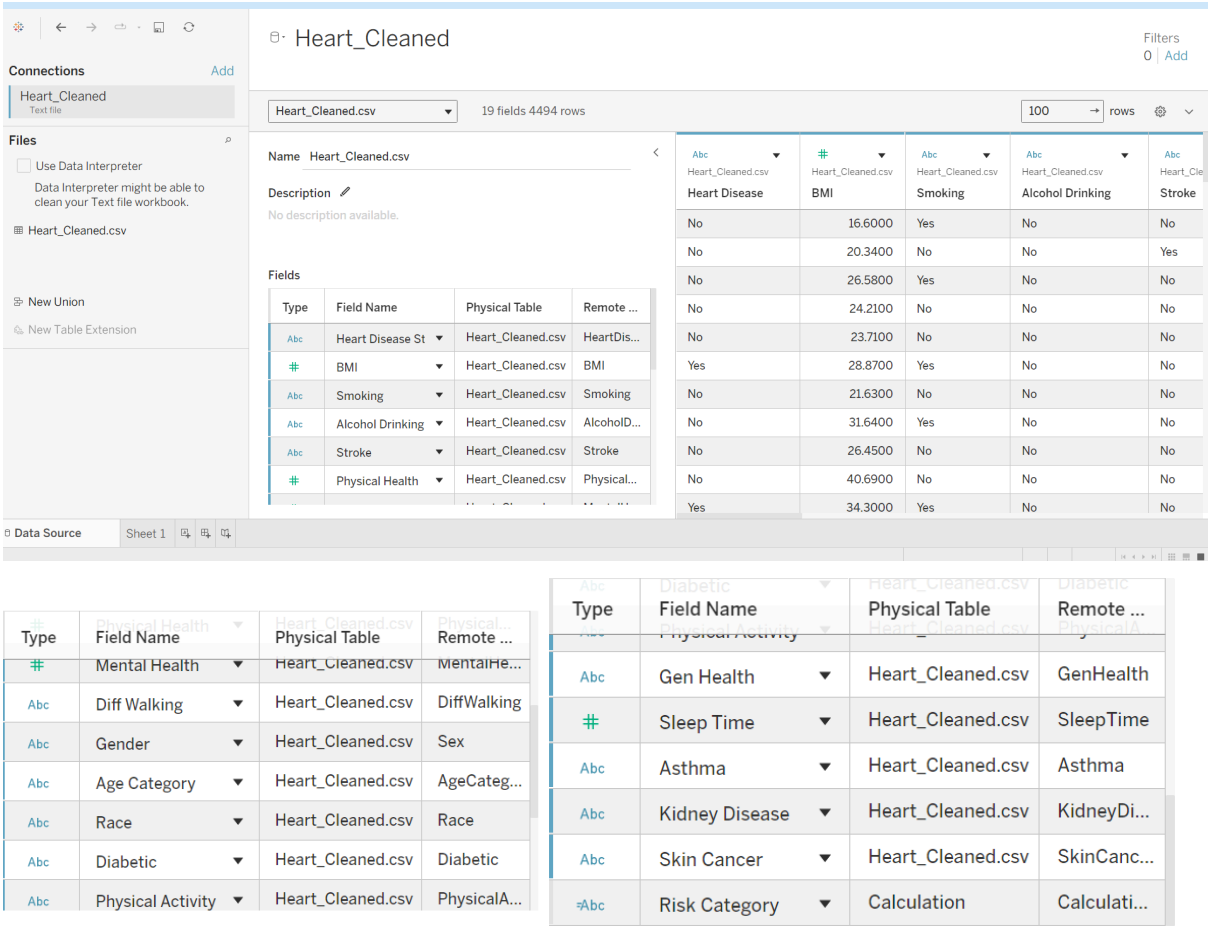
1. Introduction

In this stage of the project, the dataset was prepared for visualization in Tableau. Although the dataset was pre-cleaned, a structured review process was conducted to ensure it was fully ready for analysis and dashboard development.

2. Data Review & Exploration



The dataset was imported into Tableau and reviewed for:

- Data types (dimensions and measures)

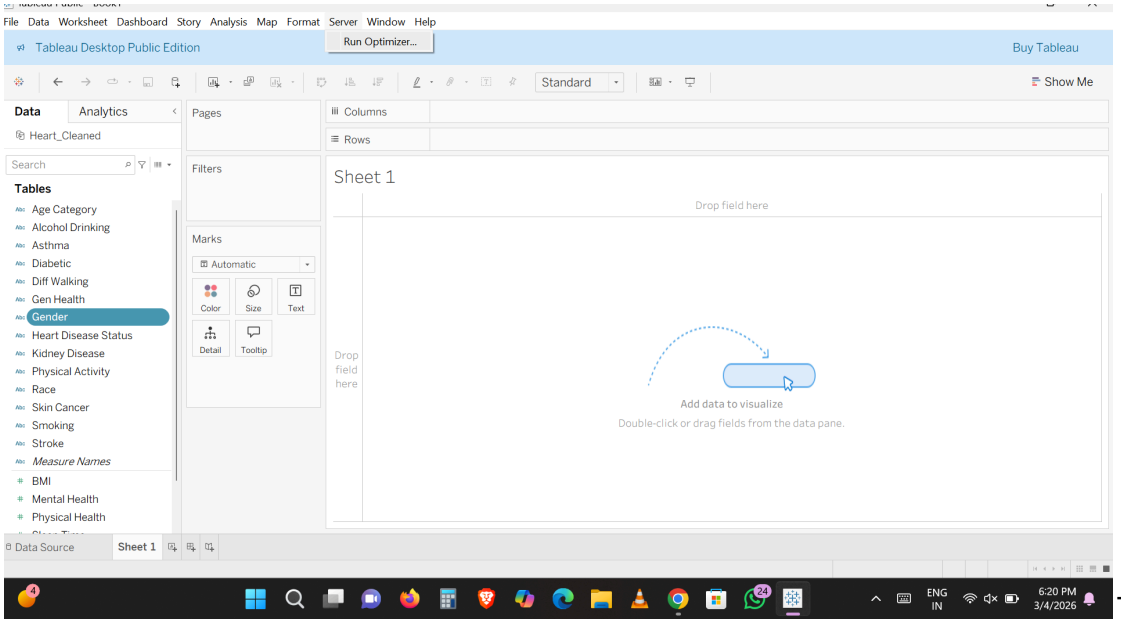
- Field consistency
- Value distributions
- Record counts

Key observations:

- Categorical fields such as Gender, Smoking, and Heart Disease Status were correctly identified as dimensions.
- Numeric fields such as BMI and Physical Health were verified as measures.
- No missing or inconsistent values were detected.
- Record counts were validated to ensure completeness.

This step ensured that the dataset structure was accurate and ready for analysis.

3. Field Renaming & Formatting



To improve clarity and readability in Tableau:

- “Heart Disease” was renamed to **Heart Disease Status**
- Field names were reviewed for consistency and professionalism
- Yes/No categorical formatting was verified

This helped make the dataset more presentation-ready and user-friendly.

4. Filtering & Structuring

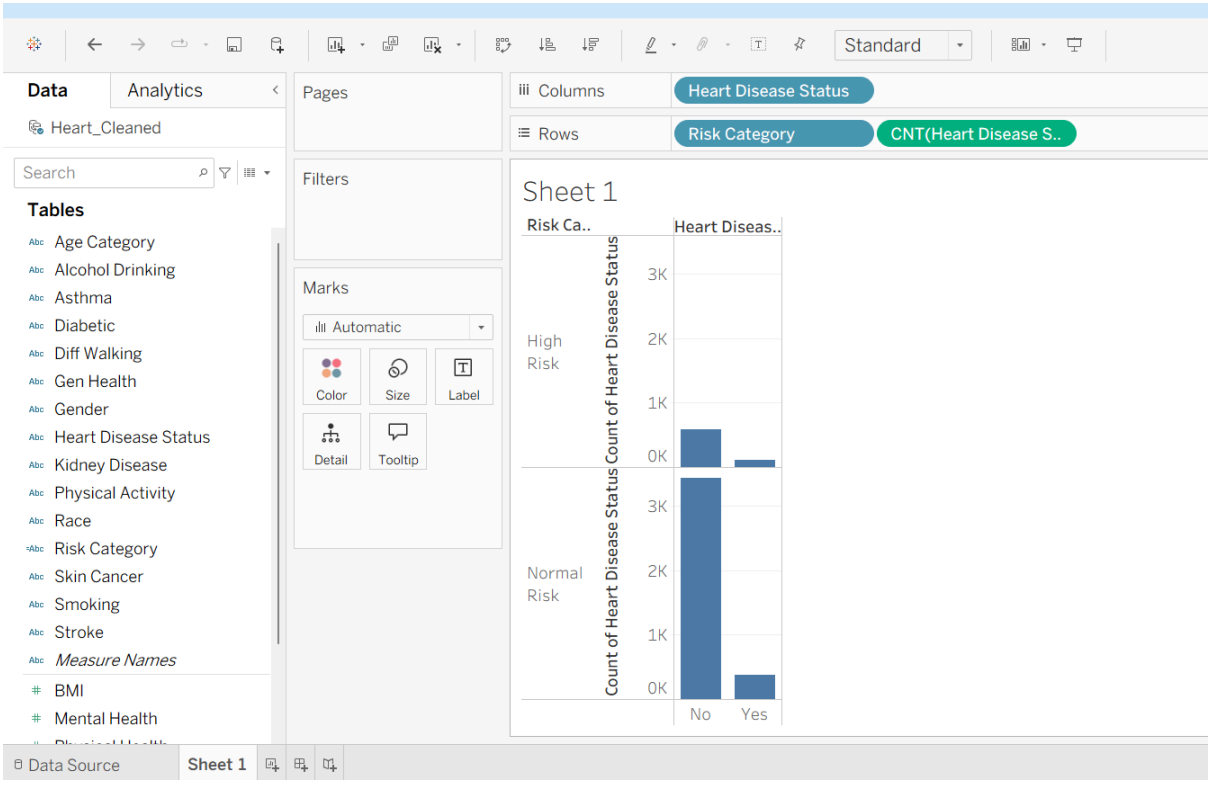
Basic filtering was tested to ensure the dataset could be segmented effectively based on analytical needs.

Example:

- Gender-based filtering was applied to verify segmentation capability.

This confirmed that the dataset is structured properly for business-focused visualizations.

5. Calculated Field Creation



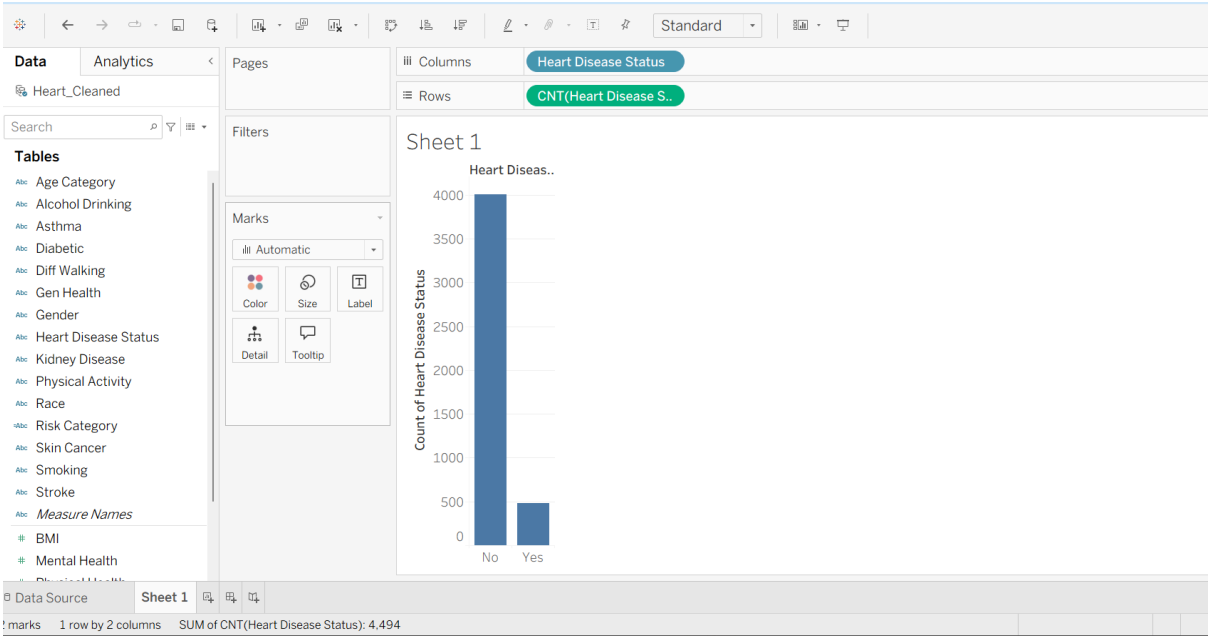
To enhance analytical capability, a calculated field named **Risk Category** was created using the following logic:

- Individuals who smoke and have BMI greater than 30 were categorized as “High Risk”

- All others were categorized as “Normal Risk”

This calculated field allows deeper insight during visualization.

6. Validation for Accuracy



Validation was performed by:

- Checking total record count
- Verifying the distribution of Heart Disease Status (Yes/No)
- Ensuring calculated field results were logically consistent

All values were confirmed to be accurate and reliable for visualization.

7. Conclusion

The dataset has been successfully reviewed, structured, formatted, and validated in Tableau. Necessary calculated fields were created, and all fields were confirmed to have appropriate data types. The dataset is now fully prepared for the Data Visualization phase of the project.

✅ **Status: Prepare the Data for Visualization – Completed**